

⚡ Nitrostack Hackathon 2026 | Healthtech & Open Innovation

SynapseMed

HEALTHCARE CONTEXT INTELLIGENCE ENGINE (HCIE) V0.3.1

Optimized Agentic Orchestration for Medical Emergencies. Connecting unstructured requests to secure, open-standard knowledge structures via the Model Context Protocol.

LANGUAGE / RUNTIME

Python 3.11+ / MCP-Native

ARCHITECTURE STATUS

Optimized Blueprint v0.3.1

PROJECT SANDBOX

Open-Source Core

01 / Vision & First Principles

The conceptual shift from basic resource retrieval to clinical-grade medical context synthesis.

First Principle: Context, Not Diagnosis



The Challenge

Healthcare systems struggle with extremely noisy, non-standardized emergency requests and highly fragmented, legacy EHR systems.

AI models face severe hallucination risks when acting as clinical classifiers or making unsupported treatment suggestions.



The HCIE Solution

Context synthesis, not medical diagnosis. HCIE operates between unstructured requests and structured medical networks to build optimal context graphs.

It provides responders with secure execution plans and clear rationale—while keeping medicine strictly with clinicians.

Architectural Evolution (v0.3.1)

Architectural Dimension	v0.3 Legacy Pipeline	v0.3.1 Optimized Pipeline	Why It Matters (Impact)
Agent Hierarchy	9 Wrapper Agents (high latency hops)	2 Core Agents (Chief & Execution)	Collapses redundant API hops; isolates execution authority.
Discovery Flow	Serial/Sequential Chain calls	Async/Concurrent Parallel dispatch	Discovery latency is bound to max() instead of sum().
File Placement	Orphaned top-level folders	Deep integration (Context & Retrieval)	Strict one-concept-per-folder architecture.
Est. File Footprint	~85-95 source files	~78-85 source files	Easier to validate, compile, and maintain on servers.

Input Transformation Workflow

Extracting Intent, Guiding Action

- ➔ **Unstructured Query:** Raw symptom logs, coordinates, or emergency dispatch calls.
- ✂ **Machine-Operable Intent:** Translates symptoms to emergency categories with rules-based severity hints.
- 📍 **Safe Execution Plan:** Proposes action points (e.g. notify family, route ambulance, retrieve EHR records).
- 👤 **No Diagnostic Spills:** Never outputs medication dosages or disease conclusions.

LIVE BLUEPRINT PAYLOAD

```
{ "task": "emergency_orchestration", "emergency_type":  
  "cardiac_suspected", "patient_name": "John Doe",  
  "symptoms": ["chest_pain", "unconscious"], "location":  
  "Kakkanad", "severity_hint": "high" }
```

02 / Core Architecture & MCP

How the Model Context Protocol (MCP) bridges the gap between LLMs and sensitive medical records.

Three Coordination Zones



1. Client Layer

Consists of standard clinician interfaces, Claude Desktop, or custom responder frontends that speak standard JSON-RPC/SSE.



2. Secure Gateway

Auth validation, capability discovery, tool route parsing, logging, and execution locks. Contains the human-in-the-loop approval gate.



3. Context Engine

Calculates the Context Intelligence Score, shapes the reasoning graph, and outputs structured, validated, safe clinical resource targets.

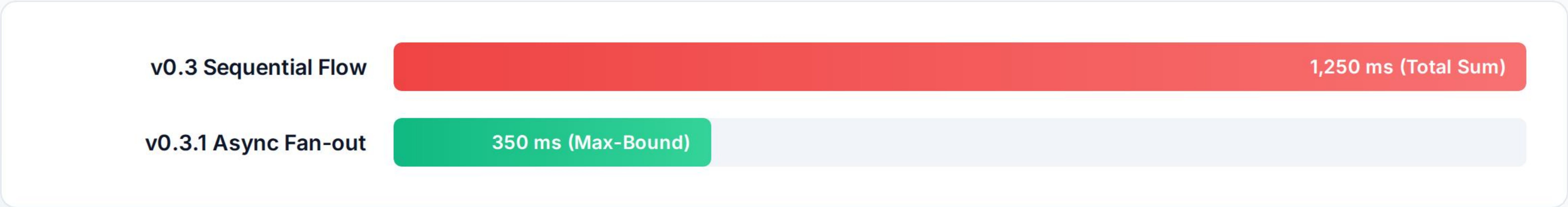
The 7-Layer Orchestration Engine

-  **Layer 1: Intent Understanding** Parses raw text into strict classifications and severity levels.
-  **Layer 2: Async Planning** Generates the query execution graph.
-  **Layer 3: Knowledge Discovery** Dispatches concurrent requests to independent MCP nodes.
-  **Layer 4: Hybrid Retrieval** Blends semantic embeddings and BM25 scores.
-  **Layer 5: Context Graph** Assembles connected entities (Patient → Doctor → Hospital) inside memory.
-  **Layer 6: Prompt Generation** Generates context-dense LLM instructions.
-  **Layer 7: Guardrail Validation** Runs strict clinical validation; blocks non-compliant outputs.

Discovery Latency Optimization

Execution Flow Metrics: Sequential vs. Parallel (Concurrent)

Under the v0.3.1 design, four independent lookup threads run asynchronously, bounding overall latency to the slowest individual service.



Optimized with `asyncio.gather` , slashing API transaction bottlenecks by up to 72%.

Algorithmic Infrastructure

Context Intelligence Score

Dynamically ranks candidates by synthesizing text similarity, department availability, geographical proximity, and operational status.

MathML Implementation Core:

$$\text{CIS} = 0.35S_{\text{sim}} + 0.25P_{\text{prox}} + 0.20D_{\text{dept}} + 0.10R_{\text{res}}$$

Priority & Inference Logic

Translates patient history & current symptoms into routing rules before prompt construction (e.g., matching prior cardiac issues directly to ICU units).

```
def infer_priority_resources(context_graph): priorities = []  
if has_symptom("chest_pain") and has_history("heart_disease"):  
    priorities.append("cardiology_priority") if  
has_symptom("unconscious"): priorities.append("icu_priority")  
return priorities
```


Future Ecosystem Expansion



Biometric Edge

Smart watch compatibility for real-time telemetry (acute heart rate spikes, fall detection) to automatically trigger emergency context assembly.



Logistics Hub

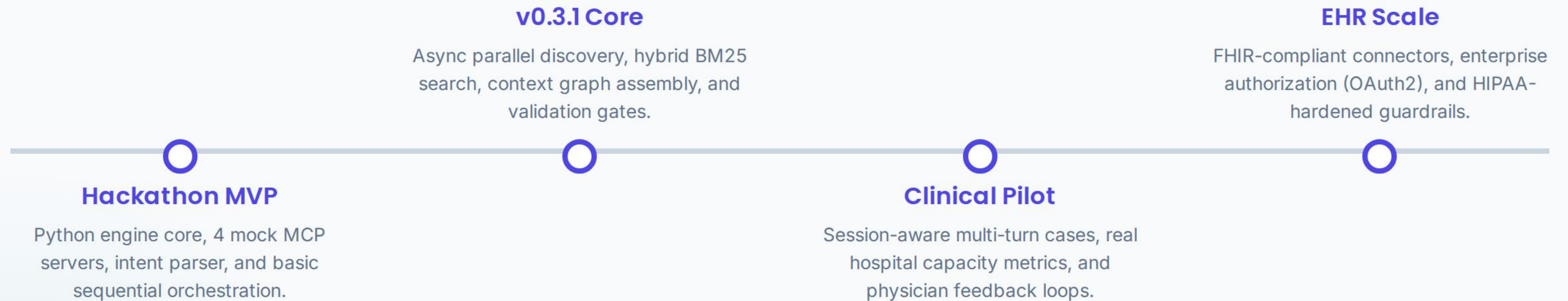
Extending open MCP server endpoints to medicine delivery systems, automated drone dispatches, and responsive ambulance marketplaces.



Clinical EHR Integration

Replacing mock JSON files with secure, production-grade HL7/FHIR gateways mapping actual electronic medical records behind stable tool schemas.

The Engineering Roadmap



 Secure Open Innovation for Healthtech

Orchestrating Healthcare Context

A standards-based, MCP-native approach to connecting LLMs safely to emergency records without medical diagnostic leakage.

GITHUB REPOSITORY

 github.com/open-hcie/core

CONTACT / CORE TEAM

 build@hcie.dirmacs.org

Disclaimer: All database items, physicians, and hospitals used in current v0.3.1 are mock-simulated data. HCIE is an orchestration proxy framework and does not provide active medical diagnoses or clinical triage.